Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI Platform · Agentic Workflows · LLM Evaluation · Python · LangGraph · PHP · FastAPI · PostgreSQL · Observability · AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years in production systems, now specializing in AI platform engineering. Built AgentProbe: open-source eval + observability infrastructure with trace-level tracing, 5-metric scoring engine, LLM-as-judge quality feedback loops, and CI-integrated regression harness — the exact charter of Cloudbeds' Core AI Squad. Built LeaseI: 8-agent LangGraph platform with durable state management, tool orchestration, fault-tolerant recovery paths, and HITL escalation (93.6% precision, 87.9% F1 across 1,240 test cases). At Entrata (hospitality-adjacent PropTech SaaS), owned PHP + Python production systems serving 100K+ monthly active users across 120+ Agile sprints at 99.5% uptime — directly applicable to Cloudbeds' PHP ecosystem. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, agentic pipelines, and data APIs; integrating AI components with backend and reporting layers.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); automated quality checks (Isolation Forest); deployed inference API (PyTorch + HuggingFace). Delivered Python and database workshops to 600+ students.

Entrata (Hospitality-Adjacent PropTech SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Owned production PHP + Python backend systems** serving 100K+ monthly active users across property management, reservations, resident communications, financial settlement, and document workflows — the same SaaS operational surface Cloudbeds powers for hotels. 120+ Agile sprints, zero compliance violations, 99.5% uptime across 7 years of full-lifecycle ownership.
- **Built fault-tolerant event-driven agentic processing** (RabbitMQ) — retry logic, idempotency guarantees, state management across thousands of concurrent account updates; owned through production incidents to permanent root-cause fixes. Designed for correctness under probabilistic, concurrent behavior.
- **Operated distributed AWS infrastructure** (S3, EC2, RDS) with observability instrumentation (OpenTelemetry, Prometheus, Grafana) — reduced debugging time by 40%; optimized 200+ SQL queries (30% gain); Docker microservices redesign (50% scalability improvement, 40% deployment time reduction). Multi-tenant PostgreSQL schema design across 100K+ units.
- Mentored 5+ engineers; 300+ peer code reviews; 80% manual operational workload eliminated via Python/Bash automation pipelines.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built PHP + Python + Java backend modules for property management workflows; led AWS S3 migration; managed data pipelines across hundreds of thousands of records.

PROJECTS

AgentProbe: AI Evaluation + Observability Platform EVAL INFRASTRUCTURE CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-judge scoring; CI regression harness with golden datasets; trace-level observability; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Directly addresses the Core AI Squad's charter: evaluate models for specific use cases, own quality feedback loops, detect regressions over time.

Leasel: 8-Agent LangGraph Platform — Agentic Harness + HITL LANGGRAPH FAULT-TOLERANT

Python · LangGraph · FastAPI · SQLite · RAG · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic harness: durable state management, tool orchestration, conditional routing, fault-tolerant HITL escalation, recovery paths, citation-grounded RAG over document corpus. 93.6% precision, 87.9% F1, pass@3 88.3% across 1,240 lease PDFs. LLM reasons; deterministic logic makes every regulated decision — correctness and reliability enforced in code, not prompts.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), PHP (7 yrs production), Java (Spring Boot), JavaScript, TypeScript, SQL, Bash — FastAPI, Django, React.js, REST APIs

AI Platform & Agentic: LangGraph, LangChain, LangSmith, Agentic Workflows, Tool Orchestration, Durable State Management, HITL, Fault-Tolerant Recovery, MCP, RAG Pipelines, ChromaDB, Hybrid Search, Embeddings, Prompt Engineering, Prompt Version Management, Guardrails, Cost Controls, OpenAI API, Anthropic API, Groq, Ollama

Evaluation & Observability: LLM Evaluation, LLM-as-Judge, Eval-Driven Development, Golden Datasets, Regression Harnesses, Trace-Level Observability, OpenTelemetry, Prometheus, Grafana, CI/CD (GitHub Actions), AgentProbe, Hallucination Detection, Scoring Methodology

Backend & Cloud: PostgreSQL, Redis, RabbitMQ, Apache Kafka, Microservices, Distributed Systems, Multi-Tenant Architecture, AWS (S3, EC2, RDS, Lambda, SQS, Step Functions, Bedrock), Docker, Kubernetes, Terraform, Event-Driven Architecture, API Design, Idempotency, Fault-Tolerant Design